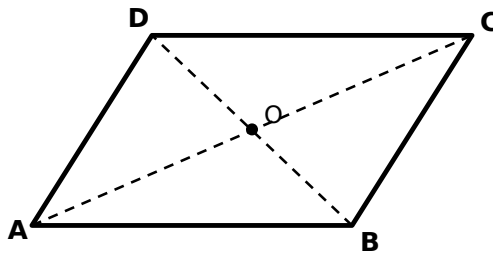


Parallelograms

A parallelogram has both pairs of opposite sides parallel. From that one fact flow all its angle and diagonal properties — the source of many exam questions.

The four properties



Parallelogram ABCD; diagonals meet at O.

1. **Opposite sides are equal** (and parallel). $AB = DC$ and $AD = BC$.
2. **Opposite angles are equal.** $\angle A = \angle C$ and $\angle B = \angle D$.
3. **Adjacent angles are supplementary** (add to 180°). $\angle A + \angle B = 180^\circ$.
4. **The diagonals bisect each other** (so $OA = OC$ and $OB = OD$). They are **not** equal in general.

Why the angles work: Because the opposite sides are parallel, a side acts as a transversal — the co-interior angles on the same side add to 180° , which makes adjacent angles supplementary and opposite angles equal.

Prerequisite refresher: "Supplementary" = two angles that add to 180° . "Adjacent angles" share a side.

Where marks slip away

Trap 1 — Opposite vs adjacent. Opposite angles are **equal**; adjacent angles are **supplementary** (add to 180°). Mixing these is the most common error.

Trap 2 — Diagonals are not equal. In a parallelogram the diagonals bisect each other but are generally **unequal** (equal only in a rectangle).

Trap 3 — Ratio of adjacent angles. If $\angle A : \angle B = 2 : 3$ and they are adjacent, then $2x + 3x = 180^\circ$, not 360° .

Trap 4 — "Diagonals bisect each other $\Rightarrow$ parallelogram." This is true and is a useful test, but bisecting at right angles makes it a rhombus.

Slip 5 — Justify with the property name ("opposite angles of a parallelogram are equal") to earn full marks.

Model answers — write them like this

Example A. In parallelogram ABCD, $\angle A : \angle B = 2 : 3$. Find $\angle A$ and $\angle B$.

Step 1: $\angle A$ and $\angle B$ are adjacent, so they are supplementary: $\angle A + \angle B = 180^\circ$.

Step 2: Let them be $2x$ and $3x$. Then $2x + 3x = 180^\circ$, so $5x = 180^\circ$, $x = 36^\circ$.

Step 3: $\angle A = 2 \times 36 = 72^\circ$, $\angle B = 3 \times 36 = 108^\circ$.

$$\therefore \angle A = 72^\circ, \angle B = 108^\circ$$

Example B. One angle of a parallelogram is three times its adjacent angle. Find all four angles.

Step 1: Let the smaller angle be x ; the other is $3x$. Adjacent angles are supplementary.

Step 2: $x + 3x = 180^\circ$, so $4x = 180^\circ$, $x = 45^\circ$.

Step 3: The angles are 45° and 135° , and opposite angles are equal.

$$\therefore 45^\circ, 135^\circ, 45^\circ, 135^\circ$$

Example C. In parallelogram ABCD, $\angle A = 70^\circ$. Find $\angle B$, $\angle C$ and $\angle D$.

Step 1: Opposite angles are equal: $\angle C = \angle A = 70^\circ$.

Step 2: Adjacent angles are supplementary: $\angle B = 180^\circ - 70^\circ = 110^\circ$.

Step 3: $\angle D = \angle B = 110^\circ$ (opposite angles).

$$\therefore \angle B = 110^\circ, \angle C = 70^\circ, \angle D = 110^\circ$$

Next: [try the worksheet](#), then [check the answer key](#).